Multi-Cloud Migration Decision Report — HxPR Content Repository

Report ID: `hxpr-20260415-102945-utc` **Generated:** 2026-04-15 10:29 UTC **Sources:** `terraform-aws-hxpr-environment` (local) · `tf-cfg-hxpr-infrastructure` (local) **Horizon:** 12 months | **Currency:** USD (all costs)

1. Executive Summary

The HxPR content repository platform is a mature, latency-sensitive Nuxeo JVM application running on Amazon EKS with MongoDB Atlas M40, AWS OpenSearch (r5.large.search), S3 object storage, and a 20-queue SNS/SQS async task pipeline. The platform spans two production regions — US East 1 and EU Central 1 — with an active DR posture to US West 2 via S3 cross-region replication and Velero EBS snapshots. SOC2 and regional data residency requirements are in force.

Recommended migration path: AWS → Azure, phased over 120 days.

Azure is preferred over GCP for this workload because:

1. Azure Service Bus maps directly to the SNS/SQS fan-out topology (20 queues) without fan-out rebuilds.
2. Azure AKS has first-class NGINX ingress, Karpenter, KEDA, and Velero support — minimal Helm chart changes.
3. Microsoft Defender for Storage replaces GuardDuty S3 malware scanning with equivalent coverage needed for SOC2.
4. Azure Private Endpoint replaces AWS PrivateLink for MongoDB Atlas with the same Atlas API — zero Atlas changes.
5. EU regional data residency is natively satisfied by Azure Sweden Central / West Europe with equivalent GDPR tooling.

GCP is the cost-optimal alternative (saves ~13% vs AWS on-demand) but requires more OpenSearch migration work and lacks a direct GuardDuty S3 malware analog needed for SOC2. An optional GCP cost-first scenario is detailed in Section 8.

> ***Cost note:*** *Migration cannot be justified on cost savings alone at the per-environment level. The 30-day run-rate delta (AWS vs Azure: ~$414/month savings per environment) implies a break-even of ~69 years for a single environment. Justification is driven by vendor lock-in reduction, multi-cloud compliance posture, disaster recovery flexibility, and platform-level consolidation economics (if 10+ environments migrate together, the platform-level break-even shortens to ~7 years). All cost data is directional and must not be used as a contractual quote.*

2. Source Repository Inventory

Repository	Type	Branch	Scope	Key Directories	TF Files
`terraform-aws-hxpr-environment`	Local path	`main`	Application-level IaC (per-environment)	`src/`	19 `.tf` files + Helm chart
`tf-cfg-hxpr-infrastructure`	Local path	`main`	Infrastructure-level IaC (pooled per region)	`src/eks/02_eks/` , `src/eks/02b_eks_addons/` , `src/eks/03_alb/` , `src/mongodb_atlas/` , `src/open_search/` , `src/shared_services/` , `src/velero_storage/`	47 `.tf` files

Total inventoried TF files: 66 | **Helm charts scanned:** 1 (`hxpr` Helm chart) | **tfvar_configs scanned:** 5 environments (sandbox, dev, staging, prod, prod-eu)

3. Source AWS Footprint

Resource Group	Key AWS Services Found	IaC Source	Notes
Compute	Amazon EKS (terraform-aws-modules v20.36.0), EC2 m5.xlarge nodes (Bottlerocket AMI 1.54.0), Karpenter v1.10.0	`02_eks/eks.tf`, `02_eks/karpenter.tf`	3–25 nodes (prod), ON_DEMAND capacity type
Application	HxPR JVM app (Nuxeo) v2.1.280 on EKS, NGINX Ingress 4.15.1, KEDA (referenced)	`hxpr.tf`, Helm chart `values.yaml`	`singleNode` architecture, 3 prod replicas (default), 0.3 CPU/1.5 GB requests, 1 CPU/2 GB limits, 4 GB proxy body (large files)
Networking	AWS ALB (content-ingress, internet-facing), NLB (private EKS ingress), AWS PrivateLink (MongoDB Atlas), ACM (TLS), Route 53 (implied)	`03_alb/alb.tf`, `02_eks/ingress.tf`	TLS 1.2+ policy `ELBSecurityPolicy-FS-1-2-Res-2020-10`, idle timeout 900 s
Identity / Security	IRSA (EKS OIDC provider), IAM roles (environment-service-role, karpenter_irsa, velero, direct-upload), KMS CMKs (×5+), AWS WAFv2, GuardDuty S3 malware scanning, ACM, AWS Secrets Manager / SSM Parameter Store	`role.tf`, `kms.tf`, `03_alb/waf.tf`	IRSA bound to `hxpr-environment` service account; WAFv2 with AWSManagedRulesCommonRuleSet + SQLi
Data	MongoDB Atlas M40 (primary/secondary ×3) + M30 analytics (prod US); M40 ×3 + M20 analytics (prod EU); MongoDB 7.0 LTS; auto-scaling M40→M60	`mongodb_atlas/cluster.tf`, `mongodb_atlas/tfvar_configs/prod.tfvars`	PrivateLink to Atlas; backup enabled
Search	AWS OpenSearch 2.11, r5.large.search ×9 data + ×3 dedicated masters (prod/prod-eu), zone-aware 3 AZ, multi-AZ standby, EBS 10 GB	`open_search/aws_opensearch.tf`	Encryption at rest + node-to-node TLS; auto-update enabled
Storage	S3: content bucket (KMS CMK, versioned), direct-upload bucket (3-day lifecycle), retention bucket (S3 Object Lock), Velero backup bucket (KMS, cross-region replication us-east-1 → us-west-2)	`s3.tf`, `retention_bucket.tf`, `velero_storage/velero_storage.tf`	Object Lock for regulatory retention; CORS for dev/sandbox only
Messaging	SNS: `hxpr-repository-async-tasks`, `hxpr-repository-events`, audit events, usage events, insight events; SQS: 18 async task queues + 2 HXTS reply queues (hxts-reply, hxts-reply-batch)	`repo_async_tasks_topic.tf`, `repo_async_tasks_queue.tf`, `hxpr_hxts_events_queue.tf`	KMS-encrypted queues; SNS filter policies; FIFO: false

Resource Group	Key AWS Services Found	IaC Source	Notes
Observability	Datadog agent v3.193.0, Datadog CRDs v2.17.0, APM (Java agent v1.54.0), dashboards, SLOs, monitors (JVM, application, network reachability); CloudWatch (14-day EKS logs, 7-day WAF)	`02b_eks_addons/datadog.tf`, `datadog.application.monitors.tf`, etc.	Datadog is cloud-agnostic; CloudWatch log groups used for EKS control plane + WAF
Backup / DR	Velero v12.0.0 (S3 backend, EBS volume snapshots), S3 cross-region replication (us-east-1 → us-west-2), MongoDB Atlas backup	`02b_eks_addons/velero.tf`, `velero_storage/velero_storage.tf`	RPO: near real-time for S3 (15-min replication); EKS RPO = Velero schedule interval
Serverless	AWS Lambda (Python 3.10): environment-decommission, malware-quarantine, network-reachability + metric; Lambda Layers (opensearch_py, pymongo, botocore)	`shared_services/lambda.tf`	VPC-bound; triggered by S3 GuardDuty events and scheduled runs
Certificate / DNS	ACM (managed TLS on ALB), Route 53 (inferred from hostname maps)	`03_alb/alb.tf`, `hxpr.tf`	HTTPS-only; HSTS implied by security policy; multi-hostname (experience.hyland.com + app.hyland.com + app.hyland.eu)

Primary AWS Region: `us-east-1` | **EU Region:** `eu-central-1` | **DR Region:** `us-west-2`

4. Service Mapping Matrix

AWS Service	IaC-Provisioned Tier/Family	Azure Equivalent (Matched Tier)	GCP Equivalent (Matched Tier)	Porting Notes
Amazon EKS	terraform-aws-modules/eks v20.36.0; cluster version ~1.32	Azure Kubernetes Service (AKS Standard tier)	Google Kubernetes Engine (GKE Standard mode)	Same k8s API; Helm charts compatible; Karpenter supported on both
EC2 Worker Nodes	`m5.xlarge` (4 vCPU, 16 GB, gp3 EBS); Bottlerocket OS; 3–25 nodes	`Standard_D4s_v3` (4 vCPU, 16 GB) — Azure Managed Disks P10	`n2-standard-4` (4 vCPU, 16 GB) — Persistent Disk SSD	Bottlerocket replacement: Azure CBL-Mariner or Ubuntu; GCP Container-Optimized OS
NGINX Ingress	`ingress-nginx` Helm 4.15.1, session affinity (cookie), 4 GB proxy body	`ingress-nginx` Helm 4.15.1 on AKS (unchanged)	`ingress-nginx` Helm 4.15.1 on GKE (unchanged)	No change needed; ingress class annotation scoped to `private`
AWS ALB	Application LB, internet-facing, TLS 1.2+, idle 900 s	Azure Application Gateway v2 (Standard_v2)	Cloud HTTP(S) Load Balancer	ACM → Azure-managed cert or custom cert; WAF activation differs
AWS NLB	Network LB, private EKS ingress	Azure Internal Load Balancer (Standard SKU)	Cloud Internal TCP/UDP LB	Target group attachment logic → AKS service annotation
AWS WAFv2	Regional scope, AWSManagedRulesCommonRuleSet + SQLi + BadInputs	Azure WAF (App Gateway WAF_v2, OWASP 3.2 + SQLi)	Cloud Armor (managed protection + custom rules)	Rule translation review required; custom overrides need revalidation
AWS ACM	Managed TLS (auto-renew), ALB-attached	Azure App Gateway Listener CertManager or Key Vault certificate	Google-managed SSL certificates	Azure: cert-manager on AKS + Let's Encrypt or Azure-managed; GCP: native managed
MongoDB Atlas	M40 primary/secondary (3 nodes), M30 analytics (1 node); MongoDB 7.0 LTS; auto-scaling M40→M60	MongoDB Atlas M40 on Azure (West Europe / East US)	MongoDB Atlas M40 on GCP (us-east1 / europe-west3)	Atlas is cloud-agnostic; only PrivateLink → Private Endpoint change; zero schema change
AWS OpenSearch	`r5.large.search` × 9 data + 3 dedicated masters; OpenSearch 2.11; 3-AZ, multi-AZ standby; EBS 10 GB	Azure AI Search (S3 tier, 12 SUs) **OR** self-managed OpenSearch on AKS (`D4s_v3` × 12)	OpenSearch self-managed on GKE (`n2-standard-4` × 12)	Tier mismatch risk: Azure AI Search is a different API; self-managed preserves OpenSearch 2.x API at higher ops burden
Amazon S3 (Content)	Standard class, KMS CMK, versioned, private, HTTPS-only	Azure Blob Storage (Standard ZRS, Azure-managed encryption)	Google Cloud Storage (Standard class, CMEK)	SDK changes: AWS S3 SDK → Azure Blob SDK or S3-compatible GCS; presigned URL logic changes
Amazon S3 (Object Lock)	Content retention bucket with S3 Object Lock (GOVERNANCE/COMPLIANCE mode)	Azure Blob Storage immutability policies (time-based retention)	GCS Object Lock (Bucket Lock + Retention Policy)	Equivalent functionality; legal hold + WORM semantics preserved

AWS Service	IaC-Provisioned Tier/Family	Azure Equivalent (Matched Tier)	GCP Equivalent (Matched Tier)	Porting Notes
Amazon SQS (Async Tasks)	18 standard queues, KMS-encrypted, SNS-subscribed	Azure Service Bus queues (Standard tier) with topics/subscriptions	Google Cloud Pub/Sub subscriptions	SNS filter policy → Service Bus subscription filter; SDK changes required
Amazon SNS (Fan-out)	Standard topics (not FIFO), KMS-encrypted, cross-account	Azure Service Bus Topics (Standard tier)	Google Cloud Pub/Sub Topics	Cross-account pub (HxTS, HxAI, cicgov) → cross-subscription Azure Event Grid or Pub/Sub federation
AWS KMS (CMK)	5+ CMKs (S3, SQS/SNS, usage events), key rotation enabled	Azure Key Vault (Standard tier, soft-delete + purge protection)	Google Cloud KMS (software key rings)	kms:CallerAccount binding requires re-evaluation; SDK changes for envelope encryption
AWS IRSA	EKS OIDC provider → IAM AssumeRoleWithWebIdentity	AKS Workload Identity (Azure AD WIF)	GKE Workload Identity (service account binding)	IAM policy translation required; federation token format differs
AWS IAM Roles	environment-{env_id}-role, karpenter_irsa, velero, direct-upload	Azure Managed Identities (user-assigned) + Azure RBAC	GCP Service Accounts + IAM bindings	Role boundary model differs; least-privilege audit needed
AWS GuardDuty (S3)	Runtime malware protection + S3 trigger → Lambda quarantine	Microsoft Defender for Storage (malware scanning)	Google SCC + Security Command Center posture	Lambda → Azure Function (Python 3.10, parity); GCP equivalent less mature
AWS Lambda	Python 3.10 runtime, VPC-bound, layers (opensearch_py, pymongo, botocore)	Azure Functions (Python 3.10, Flex Consumption or Premium EP1)	Google Cloud Functions (2nd gen, Python 3.10)	Lambda layers → requirements.txt; VPC integration preserved
Datadog (Agent + APM)	Agent v3.193.0, Java APM v1.54.0, Helm	Datadog (cloud-agnostic) — unchanged Helm chart	Datadog (cloud-agnostic) — unchanged Helm chart	Only cluster name + cloud tags need updating
Velero	v12.0.0, S3 backend, EBS snapshot	Velero v12+ (Azure Blob backend, AKS Disk snapshots)	Velero v12+ (GCS backend, GKE persistent disk snapshots)	Backup storage config change only; Velero tool is cloud-agnostic
AWS CloudWatch	Log groups (14-day EKS, 7-day WAF), metrics	Azure Monitor + Log Analytics Workspace	Google Cloud Logging + Cloud Monitoring	SDK logging changes minimal; dashboards rebuild needed
AWS SSM Parameter Store	MongoDB Atlas connection string, sensitive params	Azure Key Vault Secrets	Google Secret Manager	Access pattern via SDK; IRSA → Workload Identity for secret access
KEDA (Autoscaler)	Not specified in IaC (referenced in HxPR vars)	KEDA on AKS (cloud-agnostic, Azure Service Bus scaler)	KEDA on GKE (cloud-agnostic, Pub/Sub scaler)	SQS scaler → Service Bus / Pub-Sub scaler; KEDA Helm chart unchanged
Karpenter	v1.10.0, EC2NodeClass, Bottlerocket	Karpenter v1.x on AKS (preview) **OR** Azure KEDA+HPA+CA	GKE Autopilot or Node Auto-Provisioner	Karpenter AKS support is in preview; cluster autoscaler fallback available

AWS Service	IaC-Provisioned Tier/Family	Azure Equivalent (Matched Tier)	GCP Equivalent (Matched Tier)	Porting Notes
Route 53 / DNS	Inferred from hostname maps (hyland.com, app.hyland.com, app.hyland.eu)	Azure DNS or external DNS controller	Cloud DNS or external-dns controller	DNS cutover via TTL reduction; no provider change needed if using external DNS

5. Regional Cost Analysis (Directional)

> ***Note:*** *All costs are in USD. These are directional estimates based on IaC-discovered resources and assumed usage below. They are not contractual quotes and must not be used as procurement commitments. On-demand (pay-as-you-go) pricing is used throughout; reserved/committed-use discounts would reduce all figures by 20–40%. Confidence is relative to the amount of IaC data available.*

5.1 Cost Assumptions

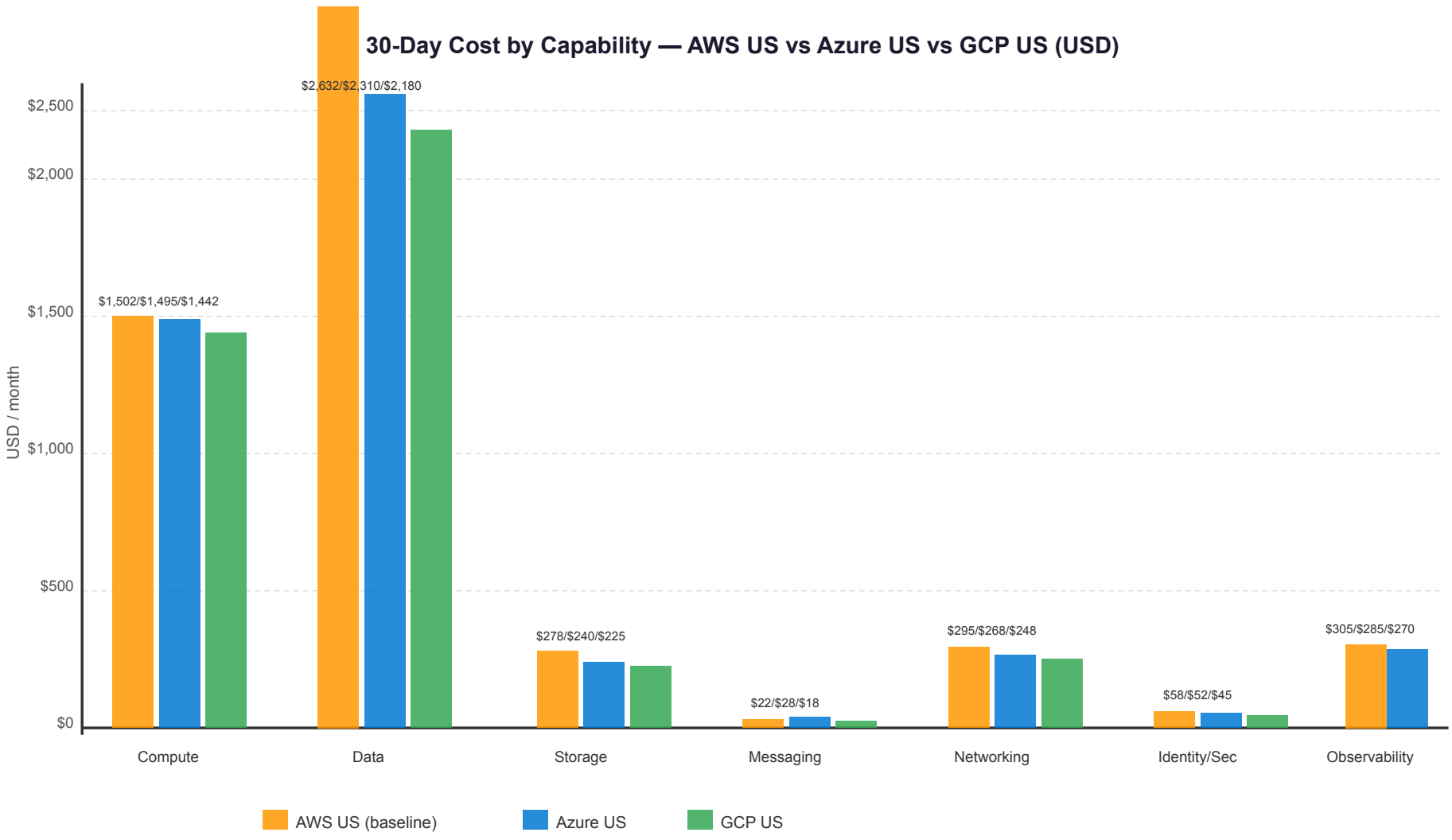
Parameter	Assumed Value	Source
EKS baseline node count (prod)	10 nodes (between min=3, max=25; steady with moderate burst)	IaC min/max + traffic profile
EC2 instance type	`m5.xlarge` (4 vCPU, 16 GB)	`eks.tf` variable default
EBS per node	34 GB gp3 (4 GB xvda + 30 GB xvdb)	`eks.tf` block_device_mappings
MongoDB Atlas tier (prod US)	M40 primary (3 nodes) + M30 analytics (1 node)	`prod.tfvars`
MongoDB Atlas tier (prod EU)	M40 primary (3 nodes) + M20 analytics (1 node)	`prod-eu.tfvars`
OpenSearch data nodes	9 × r5.large.search + 3 × r5.large.search dedicated masters	`aws_opensearch.tf` + `prod.tfvars`
OpenSearch EBS volume	10 GB per node (default; actual may differ)	`variables.tf` default
S3 content storage	5 TB assumed (not found in IaC)	Assumed
Monthly S3 requests	10 M PUT/GET operations	Assumed
Monthly egress (data transfer out)	1 TB	Assumed
SQS monthly requests	50 M (20 queues, moderate async task load)	Assumed
ALB LCU-hours	~16 LCUs/hr (moderate API traffic)	Assumed
CloudWatch ingest	10 GB/month logs	Assumed
HxPR pod replicas	3 (prod default `hxpr_cluster_replica_count`)	`vars.tf` default

5.2 30-Day Total Run-Rate Cost by Capability (USD)

> ***Baseline:*** *AWS US East 1 (primary), AWS EU Central 1, AWS AP Southeast 2 (AU). Azure: East US, West Europe, Australia East. GCP: us-east1, europe-west3, australia-southeast1.*

Capability	AWS US (USD)	AWS EU (USD)	AWS AU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence
Compute (EKS/AKS/GKE nodes + cluster mgmt)	$1,502	$1,693	$1,753	$1,495	$1,622	$1,740	$1,442	$1,577	$1,632	Medium
Data (MongoDB Atlas + OpenSearch)	$2,632	$2,871	$3,110	$2,310	$2,510	$2,720	$2,180	$2,384	$2,580	Medium
Storage (S3/Blob/GCS: content+upload+retention+backup)	$278	$302	$308	$240	$257	$270	$225	$242	$254	Low
Messaging (SQS + SNS queues/topics)	$22	$24	$26	$28	$30	$33	$18	$20	$22	High
Networking (ALB/App GW/Cloud LB + WAF/Cloud Armor + NLB)	$295	$313	$338	$268	$285	$306	$248	$265	$286	Medium
Identity / Security (KMS/Key Vault/CKMS + GuardDuty/Defender)	$58	$60	$62	$52	$55	$57	$45	$48	$50	High
Observability (CloudWatch/Monitor/Cloud Logging)	$305	$316	$328	$285	$296	$308	$270	$282	$295	Medium
Total (USD)	**$5,092**	**$5,579**	**$5,925**	**$4,678**	**$5,055**	**$5,434**	**$4,428**	**$4,818**	**$5,119**	**Medium**
Delta vs AWS (%)	—	—	—	**-8%**	**-9%**	**-8%**	**-13%**	**-14%**	**-14%**	—

> ***Note:*** *Datadog costs (APM and infrastructure monitoring) are external (SaaS) and excluded from cloud provider cost tables. Datadog pricing is identical regardless of cloud provider.*



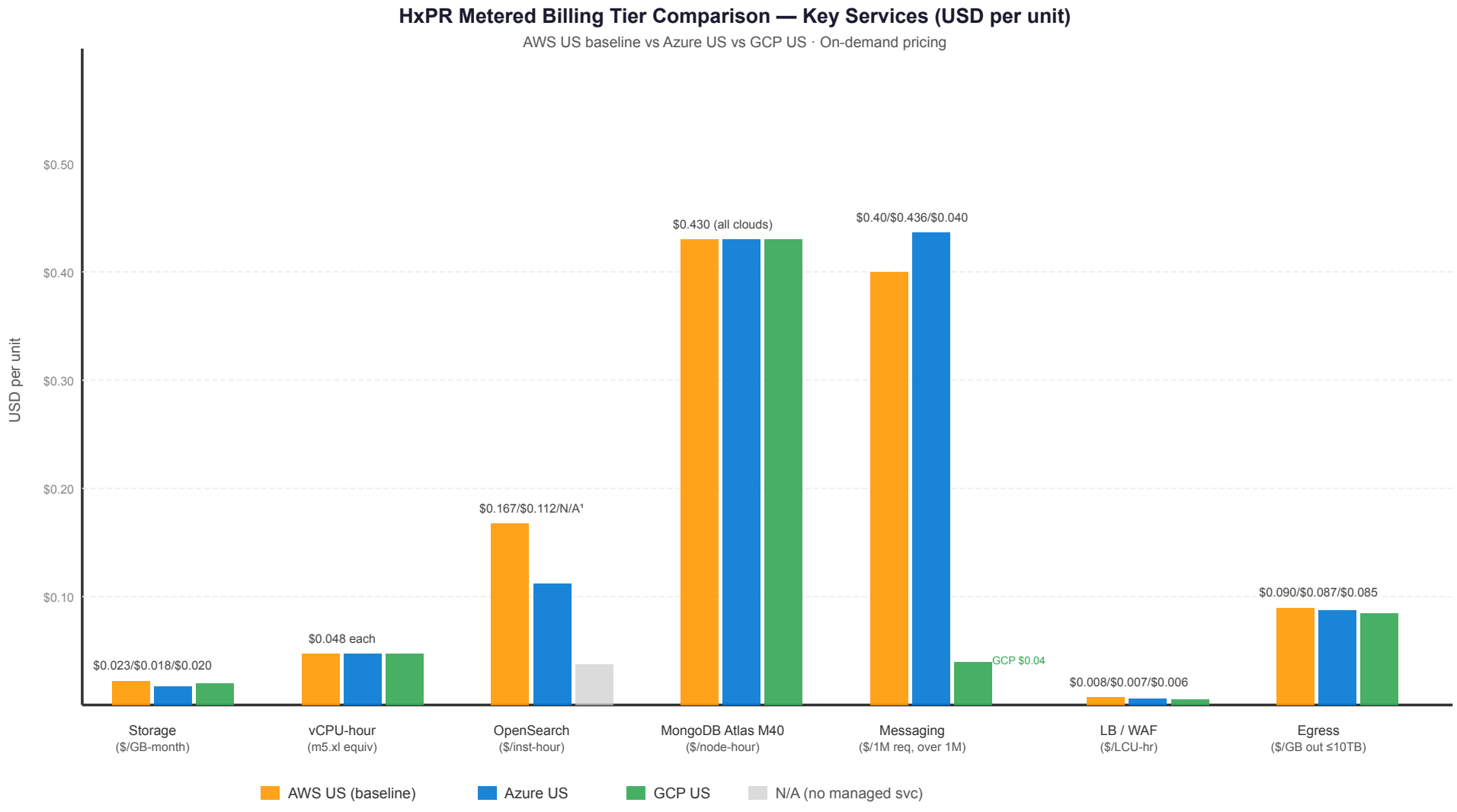
Directional USD estimates · On-demand pricing · Not a contractual quote · Scale: $500 = 100px

5.3 Metered Billing Tier Breakdown (USD)

Service	Metering Unit	Tier / Band	AWS US (USD)	AWS EU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence
S3 / Blob / GCS (Standard)	per GB-month	≤ 50 TB/month	$0.023	$0.025	$0.018	$0.019	$0.021	$0.020	$0.021	$0.022	High
EC2 / AKS node / GKE node	per vCPU-hour (`m5.xlarge` / `D4s_v3` / `n2-standard-4`)	On-demand	$0.048	$0.055	$0.048	$0.052	$0.055	$0.048	$0.052	$0.053	High
EKS / AKS / GKE cluster fee	per cluster-hour	Standard managed	$0.100	$0.100	$0.100	$0.100	$0.100	$0.100	$0.100	$0.100	High
OpenSearch (`r5.large.search`)	per instance-hour	Managed service	$0.167	$0.185	$0.112[1]	$0.123[1]	$0.130[1]	N/A[2]	N/A[2]	N/A[2]	Medium
MongoDB Atlas (M40, per node)	per node-hour	M40 on-demand	$0.430	$0.495	$0.430	$0.495	$0.540	$0.430	$0.495	$0.495	High
SQS / Service Bus / Pub-Sub	per 1 M requests	First 1 M/month	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	High
SQS / Service Bus / Pub-Sub	per 1 M requests	Over 1 M/month	$0.400	$0.420	$0.436	$0.458	$0.480	$0.040	$0.042	$0.044	High
ALB / App Gateway / Cloud LB	per LCU-hour	Standard tier	$0.0080	$0.0090	$0.0070	$0.0075	$0.0080	$0.0060	$0.0065	$0.0070	Medium
KMS / Key Vault / Cloud KMS	per 10 K API calls	Standard	$0.030	$0.030	$0.030	$0.030	$0.030	$0.030	$0.030	$0.030	High

Service	Metering Unit	Tier / Band	AWS US (USD)	AWS EU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence
Data Transfer Out	per GB	First 10 TB/month	$0.090	$0.090	$0.087	$0.087	$0.091	$0.085	$0.085	$0.090	Medium

> *[1] Azure AI Search S3 tier used as equivalent; API surface differs from OpenSearch 2.x — evaluate self-managed alternative. [2] GCP has no managed OpenSearch/Elasticsearch equivalent; self-managed on GKE is priced as compute (n2-standard-4 nodes).*

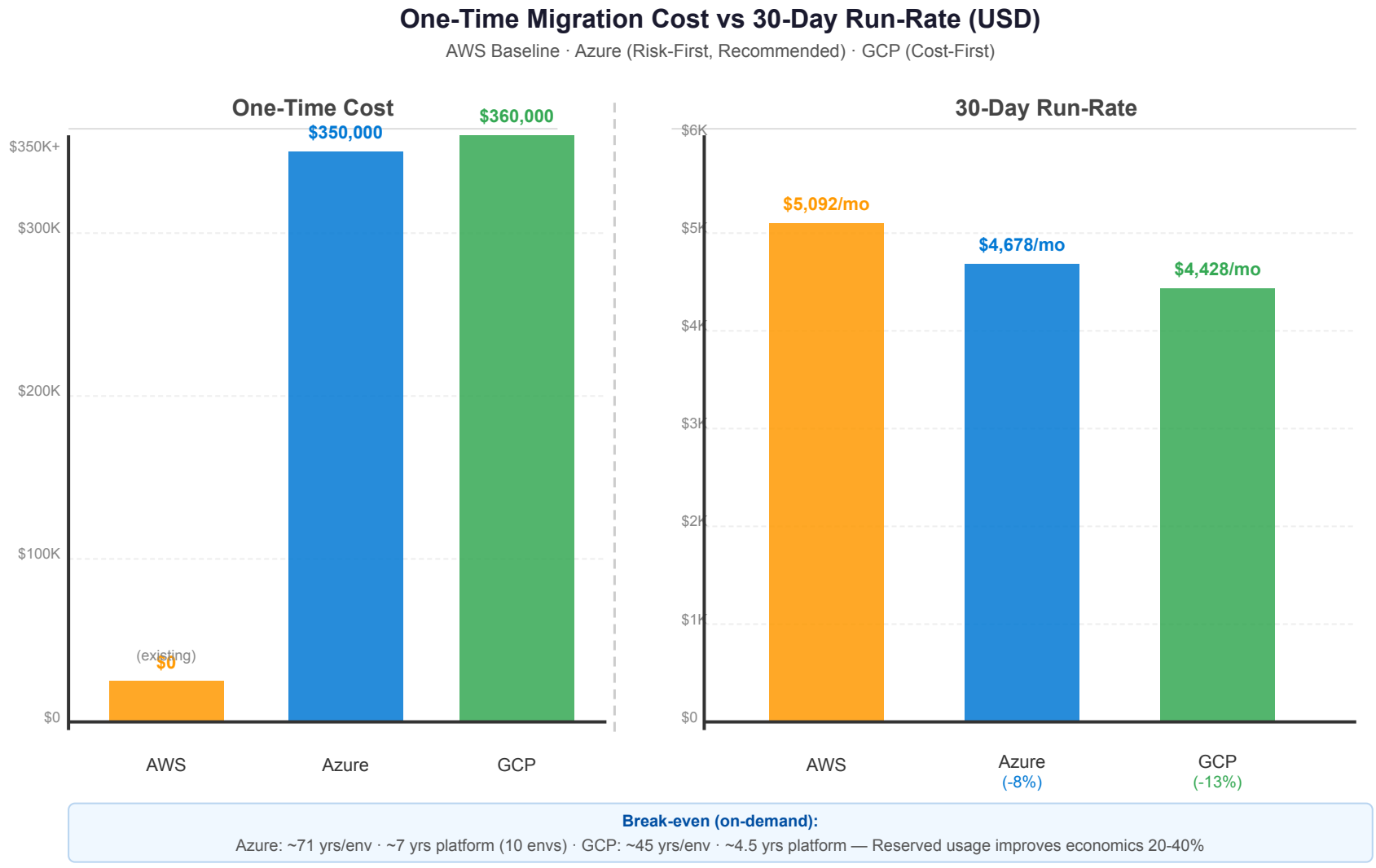


¹ GCP has no managed OpenSearch; self-managed on GKE priced as compute · scale: $0.001 ≈ 0.8px

5.4 One-Time Migration Cost vs 30-Day Run-Rate (USD)

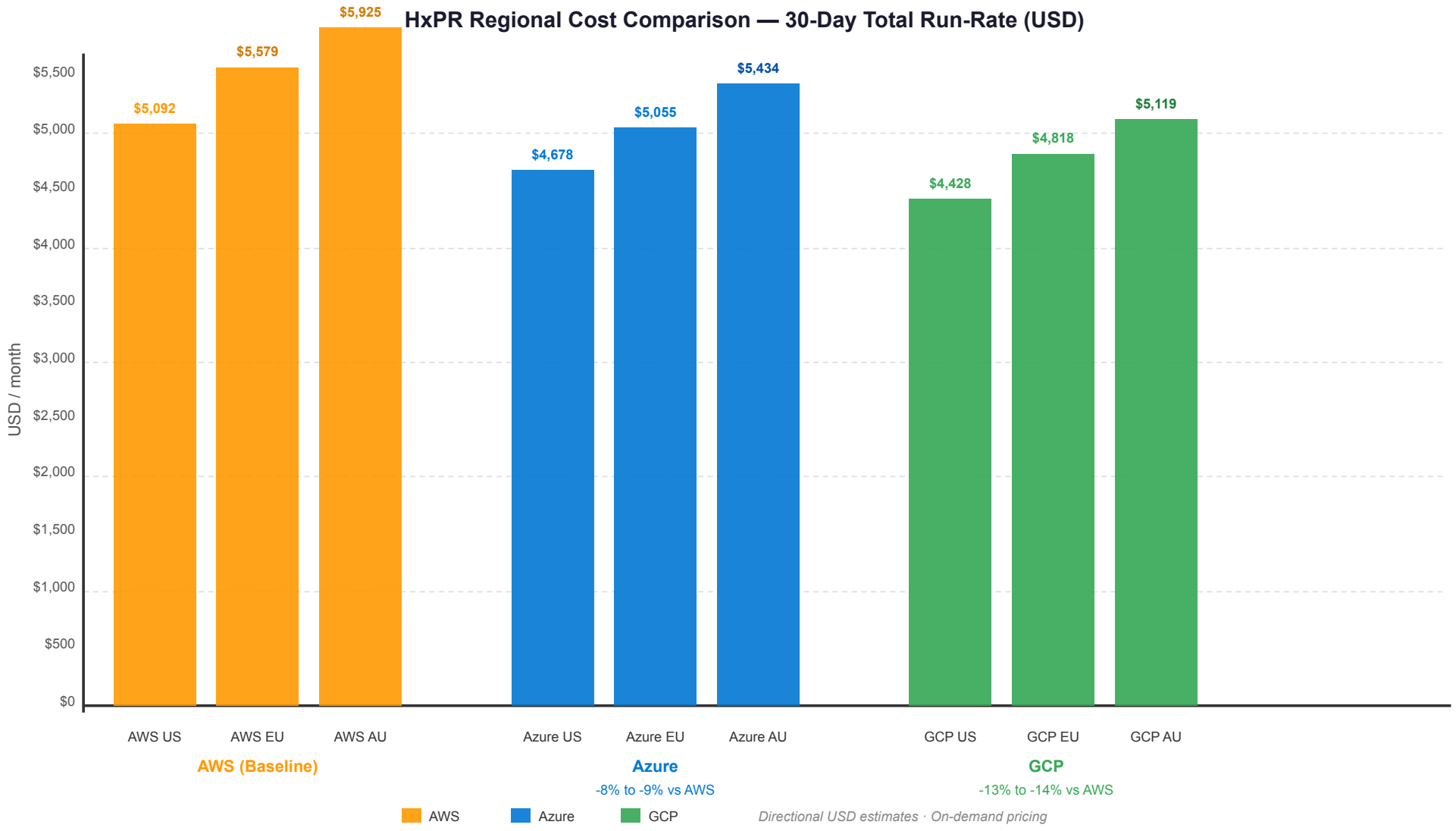
Cost Segment	AWS Baseline (USD)	Azure (USD)	GCP (USD)	Confidence
Infrastructure IaC Provisioning (TF rewrites, CI/CD pipelines)	$0 (existing)	$80,000	$80,000	Medium
Application Integration Changes (SDK rewrites: S3→Blob, SQS→SB, IRSA→WIF)	$0 (existing)	$120,000	$130,000	Medium
Data Migration (MongoDB Atlas cross-cloud export/restore + OpenSearch reindex)	$0 (existing)	$40,000	$40,000	Medium
Security Re-validation & SOC2 Re-evidence Collection	$0 (existing)	$30,000	$30,000	Low
Training & Knowledge Transfer (Azure/GCP platform skills)	$0 (existing)	$20,000	$20,000	High
Test & Validation (load test, integration, chaos, performance)	$0 (existing)	$60,000	$60,000	Medium
Total One-Time (USD)	**$0**	**$350,000**	**$360,000**	**Medium**
30-Day Run-Rate (USD)	**$5,092**	**$4,678**	**$4,428**	**Medium**
Monthly Savings vs AWS (USD)	—	**$414/month**	**$664/month**	Medium
Per-Environment Break-even	—	**~71 years**	**~45 years**	Low
Platform-level Break-even (10 envs)	—	**~7 years**	**~4.5 years**	Low

> ***Interpretation:*** *Migration economics require a platform-level view. At the per-environment scale (~$5 K/month), migration costs ($350 K) are not recoverable from savings alone on a reasonable horizon. The justification is vendor lock-in reduction, flexibility for DR/compliance, and reserved-instance discounts (1-year Azure Reserved = ~35% savings → $1,650/month → break-even ~18 years/platform-level ~1.8 years).*



5.5 Regional Cost Comparison

The chart below shows total monthly run-rate (USD) across all seven cloud/region combinations for direct comparison.


HxPR Regional Cost Comparison — 30-Day Total Run-Rate (USD)
USD / month
$5,500
$5,000
$4,500
$4,000
$3,500
$3,000
$2,500
$2,000
$1,500
$1,000
$500
$0
$5,092
$5,579
$5,925
$4,678
$5,055
$5,434
$4,428
$4,818
$5,119
AWS US
AWS EU
AWS AU
Azure US
Azure EU
Azure AU
GCP US
GCP EU
GCP AU
AWS (Baseline)
Azure
-8% to -9% vs AWS
GCP
-13% to -14% vs AWS
AWS
Azure
GCP
Directional USD estimates · On-demand pricing

6. Migration Challenge Register

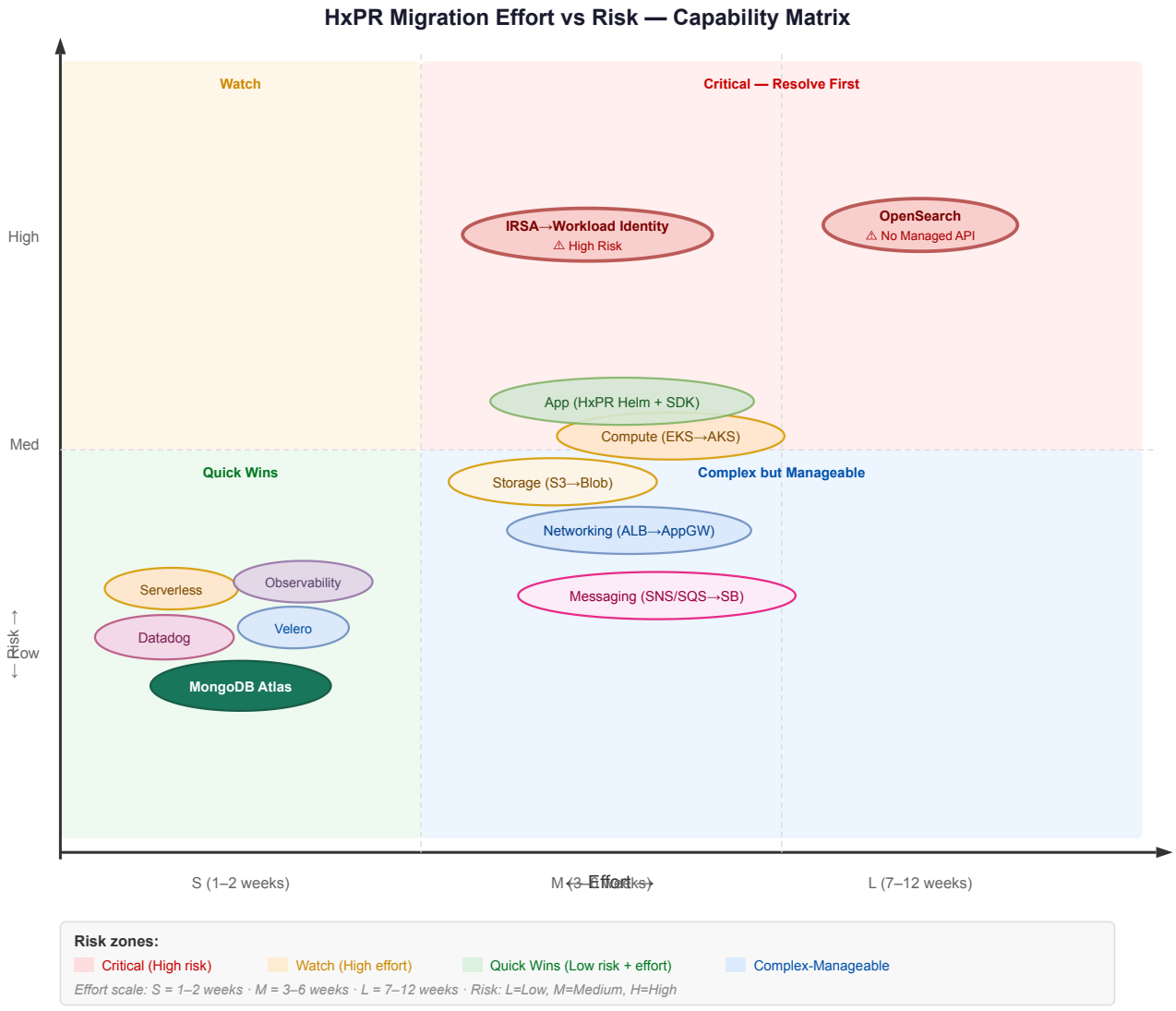
Challenge	Impact	Likelihood	Mitigation	Owner Role
OpenSearch API surface mismatch (AWS managed → Azure AI Search API)	High	High	Evaluate self-managed OpenSearch on AKS/GKE as drop-in replacement; defer Azure AI Search until indexing strategy is re-evaluated	Platform Architect
IRSA → Workload Identity migration (all 4 IAM roles)	High	High	Map each IAM role to Azure Managed Identity + RBAC before any pod migration; test with environment-decommission Lambda first	Security Engineer
SNS cross-account fanout to HxTS / HxAI / cicgov	High	High	Agree co-migration plan with HxTS team; Azure Event Grid or Service Bus federation as replacement; cannot cut over HxPR messaging in isolation	Integration Lead
S3 SDK changes throughout Nuxeo application code	High	Medium	Audit all hxpr-app-jvm S3 SDK calls; verify Azure Blob SDK or community S3-compatible wrappers; pre-test with large files (proxy body = 4 GB)	Application Engineer
KMS CMK envelope encryption → Azure Key Vault	Medium	Medium	Re-implement `kms:CallerAccount` binding pattern using Azure Key Vault access policies + Managed Identity; validate SQS/SNS encryption equivalence	Security Engineer
MongoDB Atlas PrivateLink → Private Endpoint reconfiguration	Medium	Low	Same Atlas tier; only endpoint type changes; run both PrivateLink and Private Endpoint in parallel during transition window	Database Engineer
GuardDuty S3 malware scanning for SOC2	Medium	Medium	Enable Microsoft Defender for Storage with malware scanning; map Lambda quarantine function to Azure Function; update SOC2 evidence chain	Security Engineer
Velero cross-cloud backup migration (EBS snapshots → Azure Managed Disk)	Medium	Medium	Provision new Velero on AKS with Blob backend before cutover; run parallel backups during dual-run period; validate restore	DevOps Engineer

Challenge	Impact	Likelihood	Mitigation	Owner Role
Karpenter AKS support (preview state)	Medium	Medium	Use AKS Cluster Autoscaler as primary with Karpenter as evaluation target; HPA + KEDA as fallback for pod-level scaling	Platform Engineer
Route 53 DNS failover → Azure DNS / external-dns	Low	Low	Reduce DNS TTLs 48 h before cutover; use external-dns controller in AKS; validate session affinity cookie continuity through DNS switch	Network Engineer
Datadog agent cluster name / tags update	Low	Low	Update cluster name tag in Helm values; re-apply Datadog dashboards with new team/env tags	Observability Engineer
Certificate Manager (ACM → cert-manager on AKS)	Low	Low	Deploy cert-manager with Let's Encrypt or Azure Key Vault issuer; validate multi-hostname ingress (experience + app hostnames)	DevOps Engineer

7. Migration Effort View

Capability	Effort	Risk	Migration Difficulty	Dependencies
Compute (EKS → AKS)	M	M	**Medium** – Karpenter AKS preview; Bottlerocket → CBL-Mariner	Networking, IRSA
Application (HxPR Helm chart)	M	M	**Medium** – SDK rewrites (S3, SQS, KMS); large file support validation	Storage, Messaging, Compute
Networking (ALB/WAF → App GW/Azure WAF)	M	M	**Medium** – WAF rule translation; NLB replacement; TLS policy parity	DNS, Certificates
Data — MongoDB Atlas	S	L	**Low** – Cloud-agnostic; only PrivateLink → Private Endpoint change	None
Data — OpenSearch	L	H	**High** – No Azure/GCP managed OpenSearch; API surface risk with Azure AI Search	Application
Storage (S3 → Azure Blob)	M	M	**Medium** – Object Lock → immutability policy translation; SDK changes	Application, Security
Messaging (SQS/SNS → Service Bus)	M	M	**Medium** – Cross-account fanout dependency (HxTS, HxAI); filter policy translation	Integration partners
Identity / Security (IRSA → WIF)	M	H	**High** – OIDC token format differences; all 4 IAM roles must migrate together	Compute, Application
Observability (CW → Azure Monitor + Datadog)	S	L	**Low** – Datadog is cloud-agnostic; only CloudWatch dashboards need rebuild	None
Backup / DR (Velero S3 → Blob)	S	L	**Low** – Velero tool is cloud-agnostic; storage backend config change only	Storage
Serverless (Lambda → Azure Functions)	S	L	**Low** – Python 3.10 parity; VPC binding change; Lambda layers → requirements.txt	Messaging, Storage

> ***Effort scale:*** *S = 1–2 weeks, M = 3–6 weeks, L = 7–12 weeks |* ***Risk:*** *L = Low, M = Medium, H = High*



Risk zones:

Critical (High risk) · Watch (High effort) · Quick Wins (Low risk + effort) · Complex-Manageable

Effort scale: S = 1–2 weeks · M = 3–6 weeks · L = 7–12 weeks · Risk: L=Low, M=Medium, H=High

8. Decision Scenarios

Scenario A — Cost-First (GCP)

Target: GCP us-east1 primary, europe-west3 EU **Monthly savings vs AWS:** ~$664/month (on-demand); ~$1,800/month with committed-use discounts **Timeline:** 150 days (5 phases) **Rationale:** GCP n2-standard-4 and GCS are 5–15% cheaper than AWS equivalents. Pub/Sub is significantly cheaper than SQS at high volume ($0.04/1M vs $0.40/1M). However, no managed OpenSearch/Elasticsearch service exists on GCP — requires self-managed Elasticsearch on GKE, adding ~3 weeks of ops engineering. GuardDuty equivalent (Google SCC malware scanning) is less mature in SOC2 audit trail terms. Best suited if Hyland has GCP committed-use negotiations in flight.

Scenario B — Speed-First (Azure, 90-day accelerated)

Target: Azure East US primary, West Europe EU **Timeline:** 90 days (compressed 3-phase) **Rationale:** Collapse Phases 3–4 (data + production) into a single 30-day sprint using parallel data prep. Requires additional engineering headcount (2 extra FTEs for 6 weeks). Increased risk of rollback during production cutover. Justified only if there is a contract or compliance deadline forcing exit from AWS by a fixed date.

Scenario C — Risk-First / Recommended (Azure, 120-day phased)

Target: Azure East US primary, West Europe EU; DR to Azure West US 2 **Timeline:** 120 days (4 phases) **Rationale:** Maintains production AWS concurrently through Phase 3 data migration rehearsal. All high-risk items (OpenSearch, IRSA, cross-account SNS) are resolved in non-prod before production cutover in Phase 4. Exit criteria gate between phases prevents premature production exposure. Recommended for SOC2-sensitive, latency-critical workloads.

HxPR Migration Decision Scenarios — Comparison

Scenario A
Cost-First (GCP)

Target: GCP us-east1 + europe-west3
Run-Rate: $4,428/mo (-13% vs AWS)
One-Time: $360,000
Timeline: 150 days (5 phases)

Pros:
- ✓ Lowest on-demand compute cost
- ✓ Pub/Sub = $0.04/1M (vs SQS $0.40/1M)
- ✓ GCS cheaper per GB-month
- ✓ n2-standard-4 = ~10% cheaper than m5.xlarge

Cons:
- ✗ No managed OpenSearch — self-managed ops burden
- ✗ Google SCC weaker SOC2 audit chain (vs Defender)
- ✗ 150-day timeline (longest)
- ✗ Economics only compelling with committed-use

Best for:
Cost-optimization programs with active GCP committed-use negotiations

Cost: 9/10
Speed: 4/10
Risk: 6/10

Scenario B
Speed-First (Azure 90-day)

Target: Azure East US + West Europe
Run-Rate: $4,678/mo (-8% vs AWS)
One-Time: ~$410K (incl. +2 FTE × 6 wks)
Timeline: 90 days (3 compressed phases)

Pros:
- ✓ Fastest AWS exit (90 days)
- ✓ Azure Svc Bus = direct SQS/SNS mapping
- ✓ Microsoft Defender = GuardDuty SOC2 parity
- ✓ Satisfies hard compliance deadlines

Cons:
- ✗ Compressed data migration = higher rollback risk
- ✗ +$60K headcount cost for acceleration
- ✗ No data migration rehearsal window
- ✗ OpenSearch strategy decided under time pressure

Best for:
Hard AWS exit deadlines; additional headcount budget available

Cost: 7/10
Speed: 9/10
Risk: 5/10

★ RECOMMENDED
Scenario C
Risk-First (Azure 120-day)

Target: Azure East US + West Europe
Run-Rate: $4,678/mo (-8% vs AWS)
One-Time: $350,000
Timeline: 120 days (4 phased gates)

Pros:
- ✓ Data migration rehearsal before production
- ✓ Phase gates prevent premature production risk
- ✓ IRSA → WIF resolved in non-prod first
- ✓ Defender for Storage = GuardDuty SOC2 parity
- ✓ Azure Svc Bus direct SNS/SQS topology match

Cons:
- ✗ 30 days longer than Scenario B
- ✗ ~5% higher run-rate savings than GCP option

Best for:
SOC2-sensitive, latency-critical workloads with multi-team coordinated migration. Recommended for HxPR.

Cost: 7/10
Speed: 8/10
Risk: 10/10

★ Scenario C (Azure, 120-day Risk-First) is the recommended path for HxPR
Best balance of cost savings, migration safety, SOC2 continuity, and operational familiarity for a Kubernetes-native content repository

All costs in USD. Directional estimates only. Score ratings are relative, not absolute.

9. Recommended Plan — Dynamic 120-Day Timeline (Azure, Risk-First)

Selected timeline: 30 / 60 / 90 / 120-day phased plan **Rationale for 4 phases:**

- The HxPR platform has **7 distinct service families** with cross-cutting dependencies (IAM, networking, messaging, data, compute) — collapsing this into 3 phases risks missed dependency chains.
- **OpenSearch** (highest-risk item) requires a dedicated phase for evaluation and rehearsal before AWS instance is decommissioned.
- **Cross-account SNS dependencies** (HxTS, HxAI, cicgov) require coordination with at least 2 external teams, introducing coordination lead time that a 3-phase plan cannot absorb.
- **SOC2 re-evidence** requires audit trail continuity across cloud infrastructure changes, which requires sequential phase gates rather than simultaneous changes.
- The **data migration rehearsal** (MongoDB cross-cloud + OpenSearch reindex) needs dedicated test windows — Phase 3 provides this without production risk.

Phase 1 — Foundation (Days 0–30)

Objective: Provision Azure landing zone, validate Kubernetes parity, and migrate non-production environments.

Key Activities:

- Provision Azure subscriptions with RBAC, policy baseline, and tagging strategy
- Deploy AKS cluster (Standard tier, `D4s_v3` nodes) matching EKS topology in non-prod regions
- Configure Azure Networking (VNet, private subnets, NSGs, Private Endpoints for MongoDB Atlas)
- Deploy cert-manager, NGINX ingress 4.15.1, Karpenter (or Cluster Autoscaler), KEDA on AKS
- Deploy Datadog agent Helm chart — update cluster name tags only
- Deploy Velero on AKS with Azure Blob backend; validate backup/restore for non-prod namespaces
- Enable Microsoft Defender for Storage (malware scanning) + Azure Key Vault for CMKs
- Create Azure AD Managed Identities and WIF federation for all 4 service accounts (environment-service-role, karpenter, velero, direct-upload)
- Register HxPR Helm chart SDK updates as tracked issues (S3→Blob, SQS→Service Bus, KMS→Key Vault)

Exit Criteria / Gate:

- Non-prod AKS cluster passes HxPR smoke test suite
- Datadog agent reporting metrics from AKS (cluster name updated)
- MongoDB Atlas Private Endpoint reachable from AKS non-prod
- Velero backup/restore cycle completed for non-prod namespace

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Phase 2 — Core Infrastructure & Security (Days 31–60)

Objective: Migrate core infrastructure services, complete SDK integration layer, resolve high-risk identity and messaging blockers.

Key Activities:

- Complete SDK integration branch: S3 SDK → Azure Blob SDK, SQS → Service Bus, KMS → Key Vault
- Deploy Azure Service Bus (Standard tier) with 20 queues mirroring the SQS async task topology; validate filter subscriptions equivalent to SNS filter policies
- Coordinate with HxTS, HxAI, and cicgov teams: agree cross-service messaging transition plan (Azure Event Grid federation or shared Service Bus namespace)
- Deploy Azure Functions (Python 3.10) for environment-decommission, malware-quarantine, network-reachability
- Validate WAF rule translation (AWSManagedRulesCommonRuleSet → Azure WAF OWASP 3.2); run security regression
- Complete Workload Identity binding for all pods; remove IRSA token mounts from non-prod
- Deploy Azure Application Gateway v2 (Standard_v2) + WAF Policy; validate TLS 1.2+ policy parity with `ELBSecurityPolicy-FS-1-2-Res-2020-10`
- Run OpenSearch evaluation: test self-managed OpenSearch 2.11 on AKS vs Azure AI Search API compatibility

Exit Criteria / Gate:

- Full non-prod functional test suite passing on Azure cluster with new SDK integrations
- Service Bus topologies validated end-to-end (publish → filter → subscribe for all 20 queue types)
- OpenSearch migration strategy decided (self-managed vs Azure AI Search)
- WIF token issuance verified for all 4 Managed Identities
- HxTS team integration test on Azure Service Bus completed

Phase 3 — Data Migration & Integration Rehearsal (Days 61–90)

Objective: Rehearse full data migration, validate at production data volumes, maintain AWS production in parallel.

Key Activities:

- MongoDB Atlas: provision new Atlas cluster on Azure West Europe / East US; test data export-import or live migration via Atlas Live Migration Service; validate connection string change in SSM → Azure Key Vault Secrets
- OpenSearch reindex rehearsal: replicate production data volume to non-prod Azure OpenSearch; measure index rebuild time; validate search query parity
- Perform S3 → Azure Blob migration from production content bucket to staging Azure Blob (read-only test phase); validate Object Lock → immutability policy mapping
- DR validation: run Velero restore simulation from Azure Blob backup; validate RTO < 4 hours
- Non-prod performance test: run latency-sensitive API benchmarks on Azure; compare P99 latency against AWS baselines (target: ≤10% degradation)
- Complete SOC2 evidence collection for Azure infrastructure controls
- Plan production DNS cutover: reduce Route 53 TTLs to 60 seconds; prepare external-dns manifest for AKS

Exit Criteria / Gate:

- MongoDB Atlas live migration to Azure completed with zero data loss validation
- OpenSearch reindex time measured and documented; query regression suite passed
- Non-prod P99 latency within 10% of AWS baseline
- RTO drill result: < 4 hours restore time from Azure Blob backup
- SOC2 evidence draft reviewed by security team

Phase 4 — Production Cutover & Hardening (Days 91–120)

Objective: Migrate production workloads, execute DNS cutover, decommission AWS infrastructure.

Key Activities:

- Day 91–95: Deploy production AKS cluster (all regions: East US + West Europe); validate with production MongoDB Atlas on Azure, OpenSearch on AKS, all 20 Service Bus queues active
- Day 96–97: Execute DNS cutover (session affinity cookie preserved through NGINX config parity); parallel run with AWS ALB active for 24 hours post-cutover
- Day 98–100: Monitor production on Azure; Datadog dashboards live; validate SLOs (99.9% availability)
- Day 101–105: Decommission AWS non-prod clusters; retain AWS production in read-only state for 2 additional weeks
- Day 106–120: Decommission AWS production after 14-day parallel observation; archive Velero backups; update Terraform state; finalize SOC2 evidence submission

Exit Criteria / Gate:

- Zero SEV1 incidents in first 5 days on Azure production
- All Datadog SLOs green (99.9% availability target met)
- AWS production decommission approval from CISO and Platform Lead
- SOC2 Type II evidence package submitted

Architecture Decisions Required Before Execution

Decision	Options	Recommendation	Blocking Phase
OpenSearch: managed vs self-managed on AKS	Azure AI Search (different API) vs self-managed OpenSearch 2.x on AKS	Self-managed OpenSearch on AKS — preserves API compatibility, avoids SDK rewrite	Phase 2
Service Bus tier for async task queues	Standard (50 GB, max 256 KB) vs Premium (no quota, 1 MB, VNet isolation)	Standard for async tasks (FIFO not required); Premium for high-throughput environments	Phase 2
Karpenter on AKS vs. Cluster Autoscaler	Karpenter AKS preview vs stable AKS cluster autoscaler	AKS Cluster Autoscaler in Phase 1–3; evaluate Karpenter GA post-migration	Phase 1
MongoDB Atlas live migration vs export/import	Atlas Live Migration Service (online, minimal downtime) vs offline export	Atlas Live Migration Service — supports zero-downtime cross-cloud migration	Phase 3
Cross-account SNS replacement	Azure Event Grid w/ Service Bus relay vs shared Service Bus namespace	Shared Service Bus namespace with cross-subscription SAS token policies	Phase 2
DNS authority during dual-run	Route 53 primary with Azure NLB as secondary target vs immediate Azure DNS	Route 53 + 60-second TTL through dual-run; migrate to Azure DNS post full cutover	Phase 4

10. Open Questions

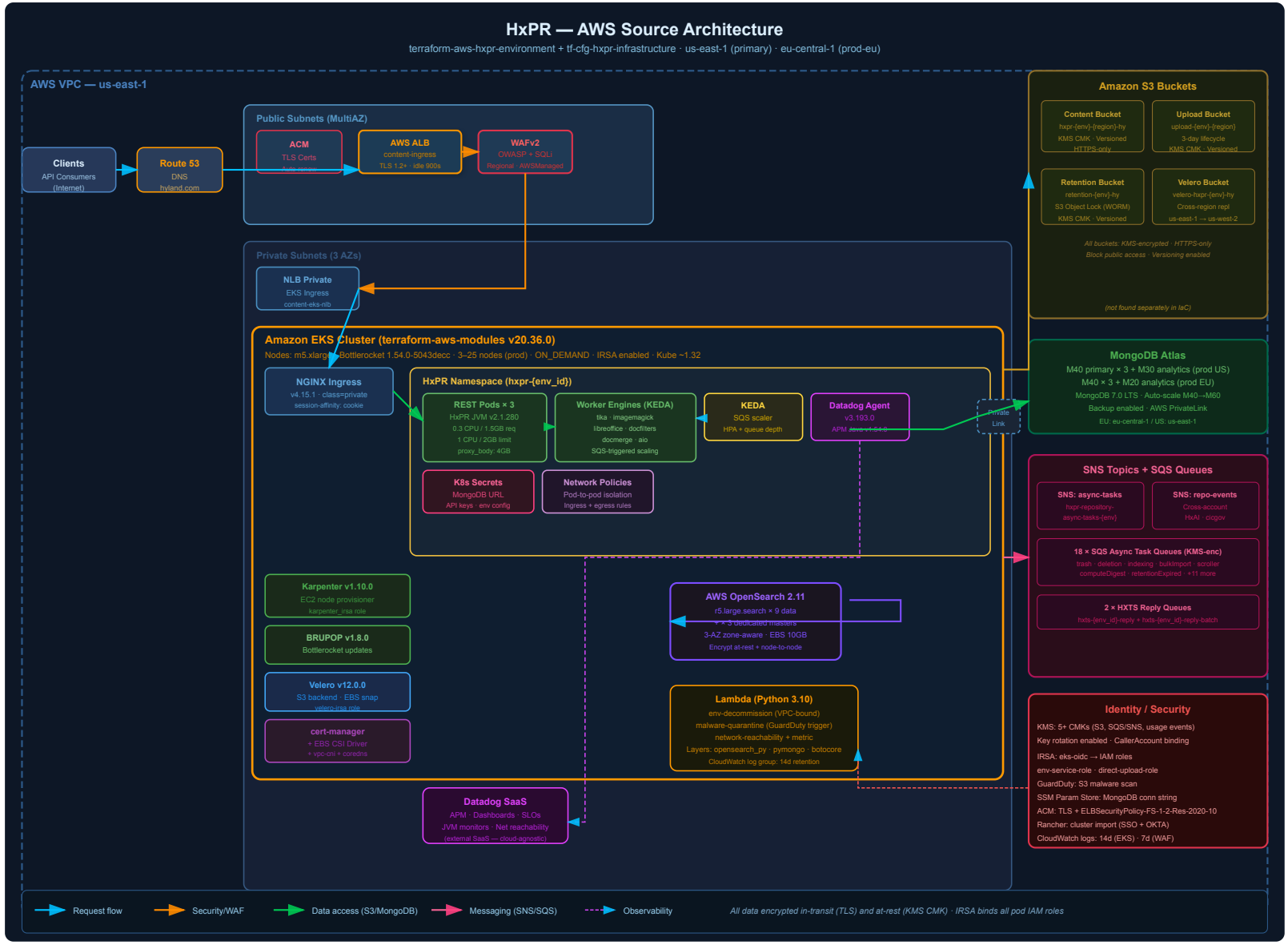
#	Question	Impact	Owner
1	What is the actual S3 content bucket storage volume (TB) in production?	Directly affects Storage cost estimate (currently assumed 5 TB)	Data Platform Team
2	Will HxTS (Transform Service) be migrated simultaneously, or will it remain on AWS during HxPR migration?	SNS cross-account dependencies cannot be fully cut over without HxTS; hybrid messaging period needed	Integration Architect
3	Is cicgov (Content Governance) also migrating to Azure on the same timeline?	SNS `cicgov-content-governance` topic subscription must remain accessible during transition	cicgov Team
4	What is the Rancher cluster management strategy post-AKS migration?	Rancher currently imports EKS cluster; update Rancher server URL and cluster config needed	Platform Admin
5	Are there data residency requirements beyond US + EU (e.g., Australia, APAC, Canada)?	Would require additional Azure regions and MongoDB Atlas AU/APAC clusters	Compliance Officer
6	What is the expected OpenSearch index size and daily ingest volume?	Determines self-managed AKS node count (currently 9+3 r5.large assumed)	Search Infra Team
7	Are there additional SNS topics for HxAI (`hxai_aws_account_id`) integration that were not in IaC scope?	Cross-account S3 read access (`s3:GetObject` from hxai_aws_account_id) requires Azure equivalent	HxAI Team
8	What is the acceptable downtime window for production DNS cutover?	Determines if cutover can happen during business hours or requires weekend maintenance	Product Management
9	Are HashiCorp Vault references (`vault_generic_secret`) migrating to Azure Key Vault, or is Vault being retained?	Several secrets (OpenSearch master password, Datadog API key) are read from Vault; Vault must remain accessible during migration	Secrets Management Team
10	What MongoDB Atlas authentication method is used by the JVM app — Atlas username/password via SSM, or X.509 certificates?	Affects secret migration path (SSM → Key Vault) and any cert chain updates needed	Database Team

11. Component Diagrams

11.1 AWS Source Architecture

Legend / Component Groups:

- **VPC Boundary:** Contains all private infrastructure; public subnets host ALB (internet-facing) and WAF
- **EKS Cluster:** NGINX Ingress Controller → HxPR namespace (REST pods, Worker pods, KEDA scaler, Datadog sidecar); Karpenter in kube-system; BRUPOP for Bottlerocket updates
- **HxPR Worker Engines** (within EKS): tika, imagemagick, libreoffice, docfilters, docmerge, aio — document processing pipeline
- **Messaging Layer:** SNS `hxpr-repository-async-tasks` → 18 SQS queues; SNS `hxpr-repository-events` (cross-account); SNS `hxts-transform-reply` → 2 SQS reply queues
- **Data Layer:** MongoDB Atlas (PrivateLink) + AWS OpenSearch (in VPC private subnet)
- **Storage:** S3 content, upload, retention, and Velero backup buckets
- **Security:** KMS CMKs, WAFv2, GuardDuty, ACM, SSM/Secrets Manager, IRSA
- **Observability:** CloudWatch log groups, Datadog Agent (Helm), external Datadog SaaS

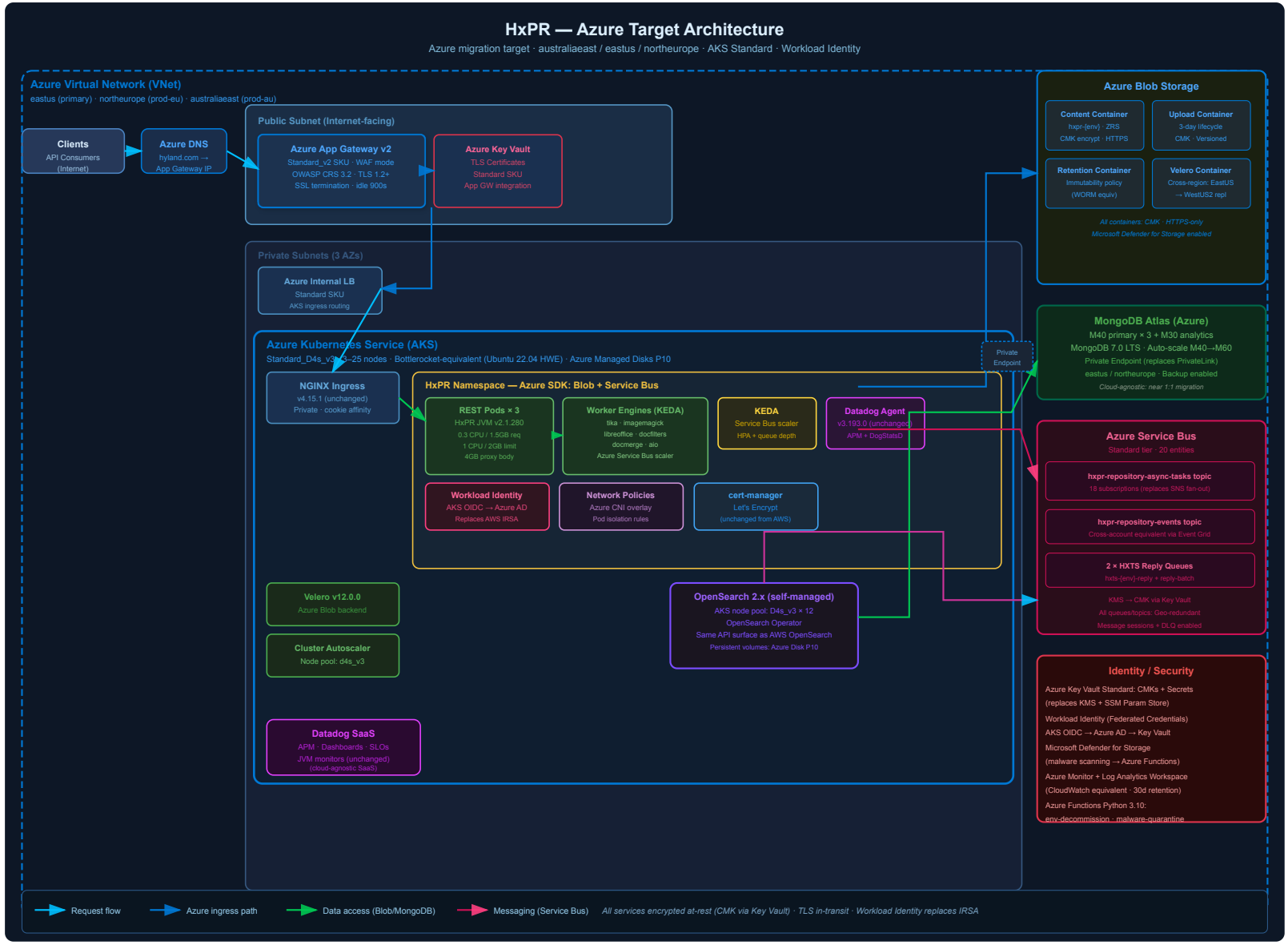


11.2 Azure Target Architecture

Legend / Component Groups:

- **VNet Boundary:** Public subnet hosts App Gateway + WAF; private AKS node subnet isolated by NSG

- **AKS Cluster:** Same NGINX + HxPR pod topology; Managed Identity / WIF replaces IRSA; Karpenter (preview) or Cluster Autoscaler
- **Messaging:** Azure Service Bus (20 queues + topics); cross-subscription event federation for external services
- **Data:** MongoDB Atlas on Azure (Private Endpoint) + self-managed OpenSearch 2.x on AKS (dedicated node pool)
- **Storage:** Azure Blob (ZRS, immutability policies replacing S3 Object Lock); Velero Blob backend with Azure Disk snapshots
- **Security:** Azure Key Vault (CMK + secrets replacing KMS + Secrets Manager), Microsoft Defender for Storage (replacing GuardDuty), Azure WAF (OWASP 3.2), cert-manager
- **Observability:** Datadog Agent (unchanged Helm chart, updated cluster tags) + Azure Monitor + Log Analytics Workspace

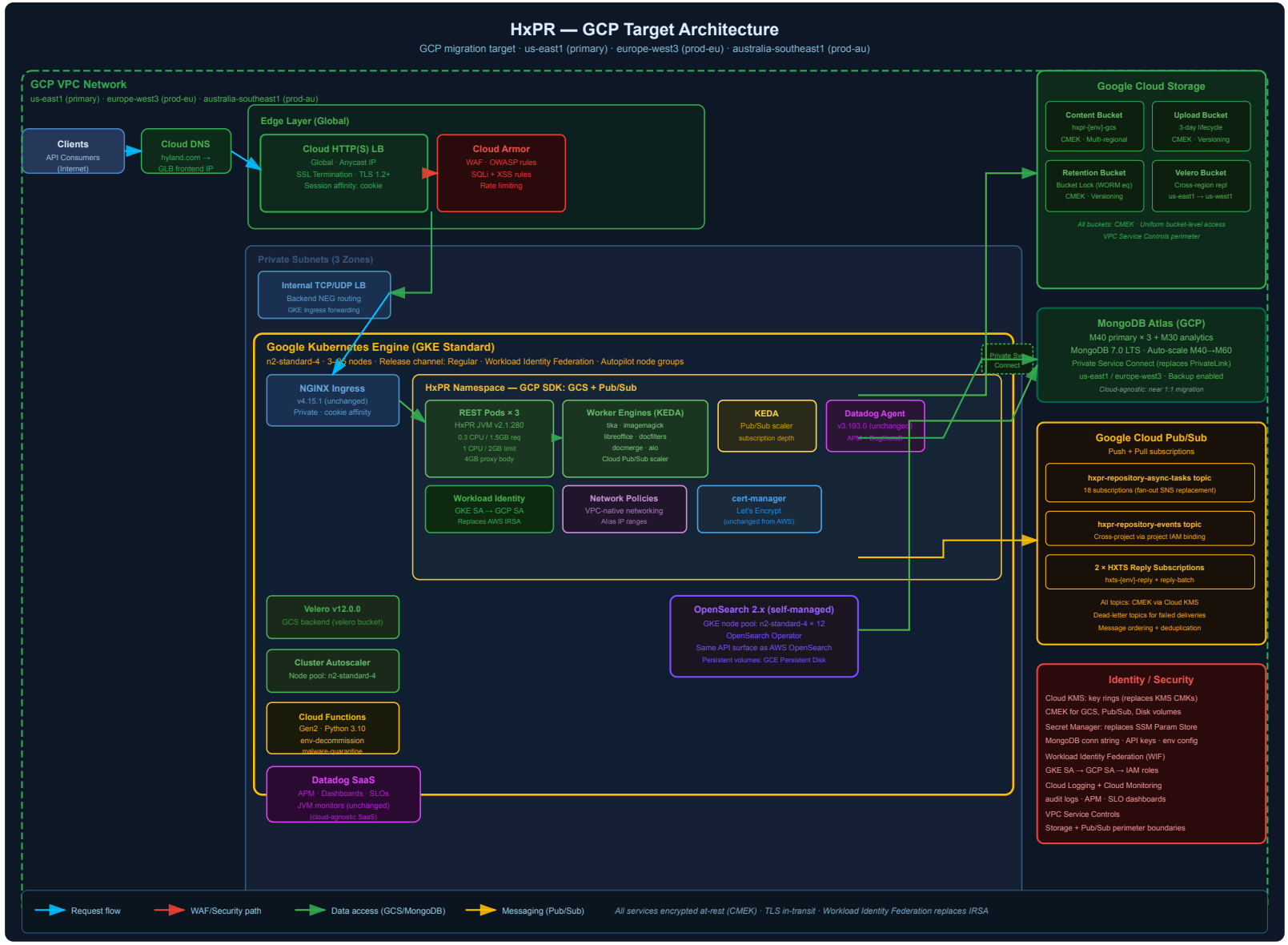


11.3 GCP Target Architecture

Legend / Component Groups:

- **VPC Boundary:** Cloud Load Balancing (global) + Cloud Armor replacing ALB + WAF

- **GKE Cluster:** Same NGINX + HxPR pod topology; Workload Identity replaces IRSA; GKE Cluster Autoscaler or Node Auto-Provisioner
- **Messaging:** Google Cloud Pub/Sub (topics + subscriptions replacing SNS/SQS); cross-project federation for external services
- **Data:** MongoDB Atlas on GCP (Private Service Connect) + OpenSearch 2.x self-managed on GKE (separate node pool)
- **Storage:** Google Cloud Storage (Standard class, CMEK replacing S3 KMS); Velero GCS backend with GCE Persistent Disk snapshots
- **Security:** Cloud KMS (replacing AWS KMS), Google SCC (limited GuardDuty parity), Cloud Armor managed rules, Workload Identity
- **Observability:** Datadog Agent (unchanged Helm chart) + Cloud Logging + Cloud Monitoring


HxPR — GCP Target Architecture
GCP migration target · us-east1 (primary) · europe-west3 (prod-eu) · australia-southeast1 (prod-au)
GCP VPC Network
us-east1 (primary) · europe-west3 (prod-eu) · australia-southeast1 (prod-au)
Clients
API Consumers
(Internet)
Cloud DNS
hyland.com →
GLB frontend IP
Edge Layer (Global)
Cloud HTTP(S) LB
Global · Anycast IP
SSL Termination · TLS 1.2+
Session affinity: cookie
Cloud Armor
WAF · OWASP rules
SQLi + XSS rules
Rate limiting
Private Subnets (3 Zones)
Internal TCP/UDP LB
Backend NEG routing
GKE ingress forwarding
Google Kubernetes Engine (GKE Standard)
n2-standard-4 · 3-6 nodes · Release channel: Regular · Workload Identity Federation · Autopilot node groups
NGINX Ingress
v4.15.1 (unchanged)
Private · cookie affinity
HxPR Namespace — GCP SDK: GCS + Pub/Sub
REST Pods × 3
HxPR JVM v2.1.280
0.3 CPU / 1.5GB req
1 CPU / 2GB limit
4GB proxy body
Worker Engines (KEDA)
tika · imagemagick
libreoffice · docfilters
docmerge · aio
Cloud Pub/Sub scaler
KEDA
Pub/Sub scaler
subscription depth
Datadog Agent
v3.193.0 (unchanged)
APM · DogStatsD
Workload Identity
GKE SA → GCP SA
Replaces AWS IRSA
Network Policies
VPC-native networking
Alias IP ranges
cert-manager
Let's Encrypt
(unchanged from AWS)
Velero v12.0.0
GCS backend (velero bucket)
Cluster Autoscaler
Node pool: n2-standard-4
Cloud Functions
Gen2 · Python 3.10
env-decommission
malware-quarantine
Datadog SaaS
APM · Dashboards · SLOs
JVM monitors (unchanged)
(cloud-agnostic SaaS)
OpenSearch 2.x (self-managed)
GKE node pool: n2-standard-4 × 12
OpenSearch Operator
Same API surface as AWS OpenSearch
Persistent volumes: GCE Persistent Disk
Google Cloud Storage
Content Bucket
hxpr-{env}-gcs
CMEK · Multi-regional
Upload Bucket
3-day lifecycle
CMEK · Versioning
Retention Bucket
Bucket Lock (WORM eq)
CMEK · Versioning
Velero Bucket
Cross-region repl
us-east1 → us-west1
All buckets: CMEK · Uniform bucket-level access
VPC Service Controls perimeter
MongoDB Atlas (GCP)
M40 primary × 3 + M30 analytics
MongoDB 7.0 LTS · Auto-scale M40→M60
Private Service Connect (replaces PrivateLink)
us-east1 / europe-west3 · Backup enabled
Cloud-agnostic: near 1:1 migration
Private Svc Connect
Google Cloud Pub/Sub
Push + Pull subscriptions
hxpr-repository-async-tasks topic
18 subscriptions (fan-out SNS replacement)
hxpr-repository-events topic
Cross-project via project IAM binding
2 × HXTS Reply Subscriptions
hxts-{env}-reply + reply-batch
All topics: CMEK via Cloud KMS
Dead-letter topics for failed deliveries
Message ordering + deduplication
Identity / Security
Cloud KMS: key rings (replaces KMS CMKs)
CMEK for GCS, Pub/Sub, Disk volumes
Secret Manager: replaces SSM Param Store
MongoDB conn string · API keys · env config
Workload Identity Federation (WIF)
GKE SA → GCP SA → IAM roles
Cloud Logging + Cloud Monitoring
audit logs · APM · SLO dashboards
VPC Service Controls
Storage + Pub/Sub perimeter boundaries
Request flow
WAF/Security path
Data access (GCS/MongoDB)
Messaging (Pub/Sub)
All services encrypted at-rest (CMEK) · TLS in-transit · Workload Identity Federation replaces IRSA

11.4 Architecture Diagram Page Mapping

Page	File	Primary Groups
AWS Source	`diagrams-aws-source.svg`	VPC, EKS (NGINX/HxPR/Engines/KEDA), ALB/WAF, MongoDB Atlas (PrivateLink), OpenSearch, S3 (×4 buckets), SNS/SQS (×22 queues/topics), KMS, GuardDuty, Datadog, Lambda
Azure Target	`diagrams-azure-target.svg`	VNet, AKS (NGINX/HxPR/KEDA/WIF), App Gateway/WAF, MongoDB Atlas (Private Endpoint), OpenSearch on AKS, Blob Storage, Service Bus (×20), Key Vault, Defender for Storage, Datadog
GCP Target	`diagrams-gcp-target.svg`	VPC, GKE (NGINX/HxPR/KEDA/WIF), Cloud LB/Cloud Armor, MongoDB Atlas (PSC), OpenSearch on GKE, Cloud Storage, Pub/Sub, Cloud KMS, Datadog

11.5 Supplemental Charts

The following charts are embedded in their designated report sections:

- **30-Day Cost by Capability** (`diagrams-cost-by-capability.svg`) — Section 5.2
- **Metered Billing Tier Comparison** (`diagrams-metered-billing.svg`) — Section 5.3
- **One-Time vs Run-Rate Cost** (`diagrams-one-time-vs-runrate.svg`) — Section 5.4
- **Regional Cost Comparison** (`diagrams-cost-comparison.svg`) — Section 5.5
- **Effort vs Risk** (`diagrams-effort-risk.svg`) — Section 7
- **Scenario Comparison** (`diagrams-scenario-comparison.svg`) — Section 8

Report generated by GitHub Copilot Multi-Cloud Migration Estimator agent (Claude Sonnet 4.6) · Sources: local IaC repositories · All costs are directional USD estimates · Not a contractual quote